

Deepak Mallampati

FORWARD DEPLOYED AI ENGINEER

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US-based (Kent, OH) - F-1 OPT, open to sponsorship & relocation

PROFESSIONAL SUMMARY

Forward-deployed AI engineer with 4+ years shipping production agentic and RAG systems end-to-end for banking, enterprise, and regulated healthcare. Builds reliable, auditable, human-in-the-loop agents that keep running when nobody is in the room - architecting a conductor-subagent orchestration framework that cut token consumption 42% while sustaining accuracy, and RAG pipelines that materially reduced hallucination over large healthcare corpora. Uses agentic coding tools (Claude Code) to ship complete production systems at frontier speed; comfortable ramping into unfamiliar codebases, integrating into someone else's stack, and earning client-engineer trust.

CORE COMPETENCIES

Agentic Systems & Orchestration • Agentic Coding Tools (Claude Code, Cursor) • RAG & Vector Retrieval • Human-in-the-Loop Reliability • End-to-End Production Delivery • Legacy Codebase Integration • Python / FastAPI • LLM Evaluation & Observability

GenAI & LLMs: GPT-4o/5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | Multi-Agent Systems | RAG Architecture | Prompt Engineering | Fine-tuning (LoRA/QLoRA/PEFT)

Retrieval & ML: FAISS, Pinecone, Weaviate | Cross-Encoder Re-ranking | Hybrid Search | PyTorch, TensorFlow | RAGAS, BERTScore

Platform & MLOps: AWS (Bedrock, SageMaker, Lambda), Azure OpenAI, GCP Vertex | Docker, Kubernetes | CI/CD | MLflow, W&B | Monitoring & Observability

Security & Compliance: HIPAA, SOC 2, PCI DSS | PII Masking | Prompt Injection Mitigation | Audit Logging

Languages: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Designed reliable, auditable, **human-in-the-loop** agent workflows with fault-tolerant batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%** - systems that keep running unattended.
- Engineered high-performance **RAG** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift relevance and materially reduce hallucination on domain-specific queries (analog to client healthcare-routing workflows).
- Established a structured **LLM evaluation & monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to surface regressions before release.
- Optimized inference cost and latency via prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Ramped into an existing internal AI platform and **integrated 25+ AI agents** and workflows into production data pipelines and observability tooling - without hand-holding on an unfamiliar stack.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Evaluated GPT-4o, Claude Sonnet, and Gemini, improving task success rates by **27%** and informing platform model-selection decisions.
- Engineered safeguards and content controls for policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining quality.
- Partnered across **20+ internal teams** to deliver scalable LLM applications supporting **100,000+ requests daily** - earning trust with engineers reviewing the work.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Built a privacy-preserving ML pipeline on clinical data (k-anonymity, l-diversity, Laplace differential privacy), cutting re-identification risk from 3.38% to 0.32% while retaining 99% of baseline predictive performance.
- Fine-tuned Qwen3-27B via 4-bit QLoRA on an H100 for controllable long-form generation; authored an IEEE-format conference paper documenting the methodology.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Built supervised regression/classification models on operational industrial data for equipment-failure prediction and anomaly detection - immersion in a domain owned by non-software engineers.
- Fine-tuned BERT/RoBERTa for entity extraction and text classification, achieving 89% F1 on custom NER tasks.

PROJECTS

career-ops

An autonomous, AI-driven job-search pipeline built and operated end-to-end with Claude Code - scans listings, scores fit, and generates tailored applications overnight without supervision. Direct proof of agentic-tool throughput: spot a workflow, ship the bespoke tool, iterate autonomously.

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites via inline DOM automation - a 0-to-1 product meeting users where they already work.

Privacy-Preserving Clinical ML Pipeline

A framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive patient data without exposing identities, quantifying the privacy-utility tradeoff per technique.

EDUCATION

M.S. in Computer Science, Kent State University, OH

2025 | GPA 3.70/4.0